LLM-brain alignment is modulated by linguistic structure even under surprisal control

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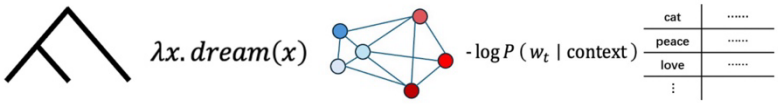
Recent proposals argue that large language models (LLMs) are compelling computational (brain-aligning) models of human language processing because LLM hidden states correlate with brain activity during language comprehension (Goldstein et al., 2022; Schrimpf et al., 2021; Tuckute, Kanwisher, et al., 2024). This view holds that the training objective of LLMs, namely next-word prediction, is also foundational for human language processing. Such a "prediction only" account contrasts with theories according to which prediction in human language processing exists alongside structural mechanisms that explain additional variance in behavioral and neural patterns above and beyond LLM-derived predictors (Huang et al., 2024; Stanojević et al., 2023). However, the two accounts are hard to disentangle since structural factors are often confounded with sequential predictability (Slaats & Martin, 2025) and these factors are likely intertwined in the daily language materials used to study LLM-brain similarity. In the current study, we adjudicate between the two accounts for brain-LLM alignment by testing the role of structural factors when predictability is controlled. We manipulate syntactic and semantic composition in English sentences while maximizing LLM-based surprisal. Then, we evaluate the linear mapping between layer-wise LLM activation and EEG recordings as a measure of alignment. Under the "prediction + structure" account, structural factors should modulate LLM-brain alignment even when predictability is constant.

We recorded 32-channel EEG while N = 35 native English-speakers read English sentences in a rapid serial visual presentation experiment. The English materials involve six conditions (**Figure 1A**): (1) **normal**: grammatical, meaningful sentences, (2) **colorless green**: sentences that follow English grammatical rules but lack a coherent meaning (3) **Jabberwocky**: sentences that only contain (inflected) English pseudowords that might cue grammatical processing, along with conditions 4-6 that are derived by **shuffling words from 1-3**. We matched conditions 2-6 in terms of surprisal (Fig 1B; based on GPT2-XL, Radford et al., 2019). Each condition has 80 experiment items, resulting in 6 * 80 = 480 critical items. 320 grammatical sentences from BLiMP were included to balance grammaticality (Warstadt et al., 2020). LLM-brain alignment is estimated with a ridge regression mapping model from LLM activations to EEG signals (Tuckute et al., 2024). The goodness-of-fit of this model quantifies the similarity between LLM and EEG. Alignment scores from the most-aligning LLM layer are then submitted to planned comparisons to test for effects of structure.

The ridge regression model yields above-chance alignment scores that outperform non-contextualized embeddings (GloVe) and random baselines. Layer 17 in the 48-layer GPT2-XL architecture was identified as the most EEG-aligning layer, and LLM-probing and EEG decoding analyses reveal that both LLM hidden layers and EEG reliably distinguish sentences along the manipulated linguistic dimensions. Planned comparisons yield three critical empirical observations: (1) Comparing alignment for shuffled and non-shuffled **Jabberwocky** sentences indicates that LLM-brain alignment *increases* in the presence of **syntax** (**Fig 2A**). (2) Comparing grammatically well-formed (but lacking a natural interpretation) **colorless green** sentences to their shuffled variants indicates that alignment *decreases* in the presence of **compositional semantics** (**Fig 2B**; independent of effects for real words compared to pseudowords). (3) **Associative semantics** does not affect LLM-brain alignment (**Fig 2C**). Our findings suggest that GPT2-XL and the brain overlap in the non-structural, *conceptual* aspect of language processing, and also that they have some shared mechanisms in terms of *syntactic* processing. Importantly, they differ in how, and potentially whether, they encode the syntax-mediated hierarchical *composition* of *semantics*. In sum, this study demonstrates how syntax, compositional semantics, and conceptual semantics modulate LLM-brain alignment when predictability is controlled.

Figure 1. (A) Example materials, design matrix and norming of (B) surprisal and (C) semantic association

A



	Syntax	Compositional Semantics	Associative Semantics	Surprisal	Lexicality
Normal sentences "Small colorful flowers bloom brightly"	✓	–	high	low	✓
Colorless green "Colorless green ideas sleep furiously"	✓	✓	low	high	✓
Jabberwocky "Rupe sleen althes rore smountly"	✓	–	low	high	–
Shuffled normal sentence "Bloom flowers colorful small brightly"	–	–	high	high	✓
Shuffled colorless green "Sleep ideas green colorless furiously"	–	–	low	high	✓
Shuffled Jabberwocky "Sleen rore rupe althes smountly"	–	–	low	high	–

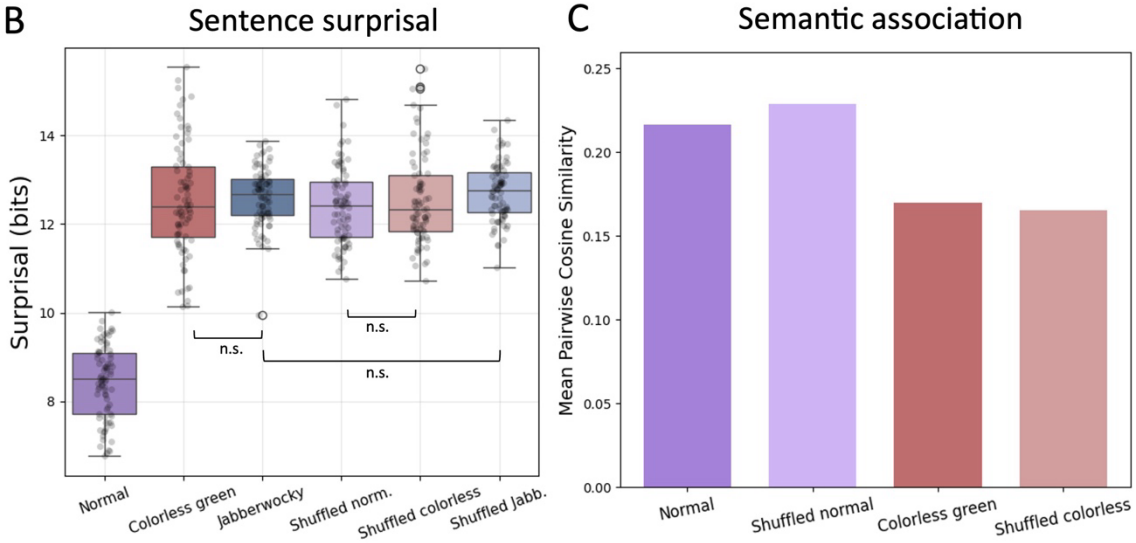
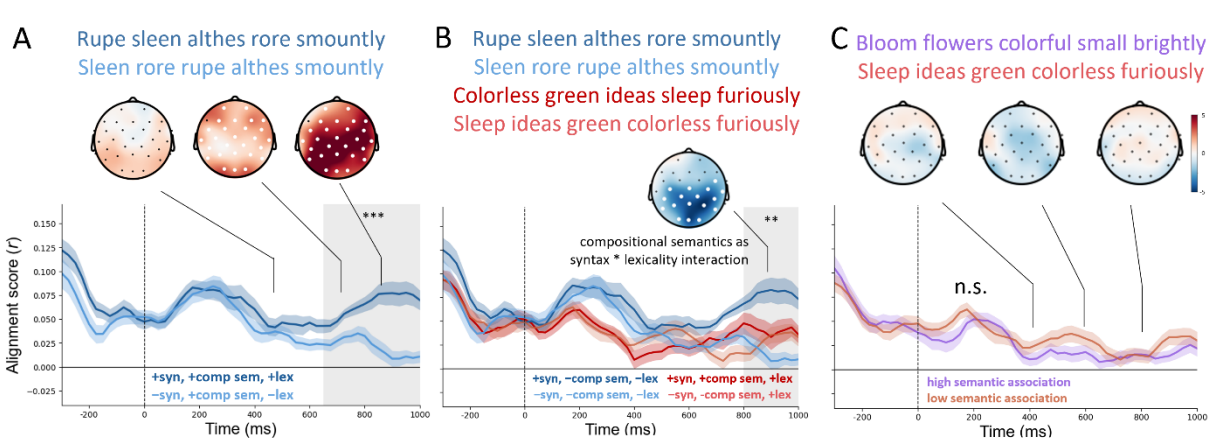



Figure 2. LLM-brain alignment comparisons for (A) syntax, (B) compositional semantics, and (C) associative semantics.